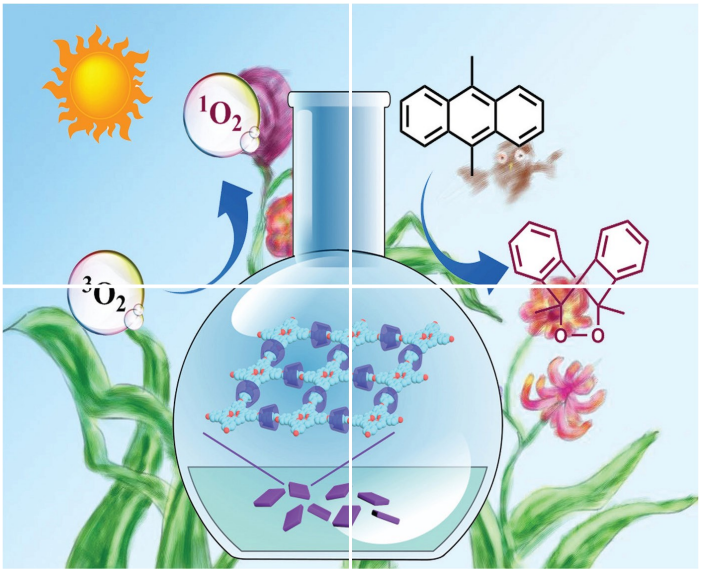
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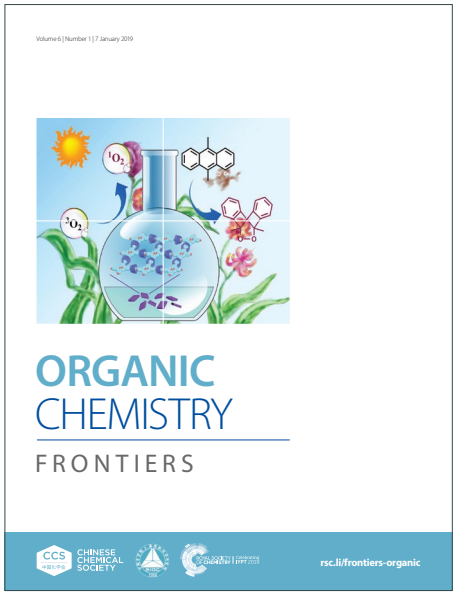
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ARTICLE

DFT study on *E*-steteoselective reductive A^3-coupling reaction of terminal alkynes with aldehydes and 3-pyrroline

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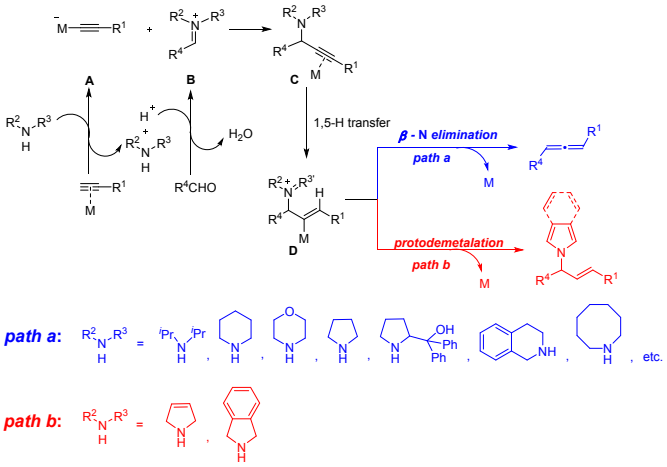
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Yuan Yao,[a] Xue Zhang*[b] and Shengming Ma*[a,b]

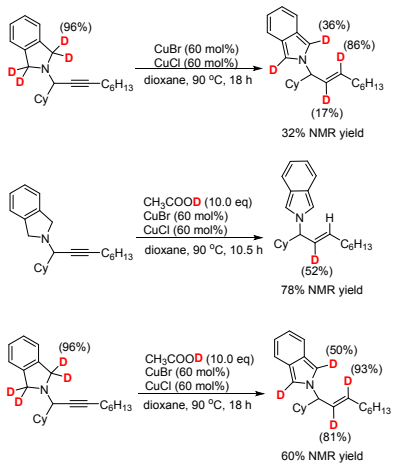
The mechanism of Cu(I)-catalyzed reductive A^3-coupling reaction of terminal alkynes with aldehydes and 3-pyrroline for synthesis *E*-allylic amines has been studied by DFT calculations. The calculations suggest that the pathway selectivity is determined by the activation energy difference between β-N elimination and protodemetalation from the iminium intermediate (**INT7**) formed *via* the rate-limiting 1,5-H transfer of the CuBr-coordinated propargylic amine (**INT4**). The stepwise intermolecular protodemetalation assisted by 3-pyrroline or water cluster proceeds more efficiently than β-N elimination, resulting in formation of allylic amine instead of allene as the final product. Notably, the aromatization of the azacycle (formed from 3-pyrroline or isoindoline) is essential driving force for the preference of the allylic amine products. In addition, the formation of hydrogen bond between the hydroxyl group of terminal alkyne and the migrating hydrogen stablizes the transition state of 1,5-H transfer step, thus, increasing the reaction reactivity.

Introduction

Metal-catalyzed three-component coupling of an alkyne, an aldehyde and an amine (A^3-coupling) is a straightforward protocol for preparing propargylic amines,[1] which serve as the key intermediates for the syntheses of many organic compounds.[2] Mechanistically, the A^3-coupling involves the reaction of the in situ generated metal alkynylide specie **A** with the iminium ion **B** *in situ* generated from the condensation of aldehyde with amine, resulting in the formation of the metal-coordinated propargylic amine **C**, which undergoes 1,5-H transfer to afford the iminium intermediate **D** (Scheme 1). Subsequent β-N elimination would produce the allene product (*path a* in Scheme 1).[3] This mechanism has been supported by our detailed theoretical studies.[3c] It's worth noting that the amine in these reactions can be diisopropylamine,[2f] dicyclohexylamine,[2g] morpholine,[2h] pyrrolidine,[2i,2k,2n] diphenylprolinol,[2j,2m] tetrahydroisoquinoline,[2l] azocane,[2o] and so on. Interestingly, *E*-allylic amines was obtained unexpectedly when terminal alkynes and aldehydes was treated with 3-pyrroline or isoindoline in the presence of copper(I) (*path b* in Scheme 1).[4] We presumed that the *E*-allylic amine is afforded through the protodemetalation process



Scheme 1. (a) A general mechanism for metal-catalyzed A^3-coupling reaction followed by 1,5-H transfer and β-N elimination affording allene (ATA reaction). (b) Reductive A^3-coupling for the synthesis of *E*-allylic amines.



Scheme 2. Deuterium-labeling experiments.

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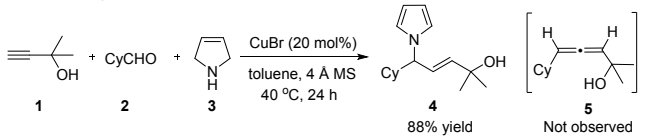
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after the generation of the iminium intermediate **D**. Deuterium-labeling experiments have been performed to confirm that the β-position proton of *E*-allylic amines came from the azacycle and the α-position proton came from the proton in ambient environment (Scheme 2).[4] Here, we sought to investigate the detailed reaction mechanism of this Cu(I)-catalyzed[5] reductive A^3-coupling reaction, as well as the crucial role of 3-pyrroline or isoindoline. The origin of the different chemoselectivity of affording allylic amine or allene was also discussed.

Computational Methods

DFT calculations were carried out with the Gaussian 09 software.[6] M06-2X,[7] including Grimme's D3 dispersion corrections, was employed to performed geometry optimizations. The LANL2DZ[8] basis set in conjunction with the LANL2DZ pseudopotential[9] was used for Cu and Br atoms, while the 6-31G(d,p)[10] basis set was used for the other atoms. Geometry optimizations were fully carried out. Harmonic vibration frequency calculations were conducted at the same level of theory to verify the stationary points to be minima (no imaginary frequency) or saddle points (one imaginary frequency). Intrinsic reaction coordinate (IRC)[11] calculations were performed to confirm the connection of the transition structures with their corresponding reactants and products. The single point energies and solvent effects in toluene (ε = 2.37) were computed with M06-2X-D3/SDD[12]-6-311++G(d,p) basis sets by using SMD[13] solvation model. The solution-phase Gibbs free energy was used in the present discussions. The calculations were carried out on the exact compounds applied experimentally: 2-methybut-3-yn-2-ol (**1**), cyclohexanaldehyde (**2**) and 3-pyrroline (**3**) with copper (I) bromide as the catalyst (Scheme 3).



Scheme 3.

Results and discussion

The A^3-coupling reaction proceeds initially from the sp C-H activation of the terminal alkyne **1** by the metal catalyst and the condensation of cyclohexanaldehyde **2** with 3-pyrroline **3**, affording the ion pair **Int3**, which consists of copper alkynylide anion **Int1** and iminium cation **Int2**. This process leads to an energy increase of 4.0 kcal mol^{-1}, accompanied by losing one molecule of water. The subsequent electrophilic attack of iminium **Int2** to copper attached C^1 *via* **TS1** requires an energy barrier of 10.1 kcal mol^{-1}, providing propargyl amine-CuBr complex **Int4**. Subsequent isomerization of **Int4** to **Int5**, affording the precursor for 1,5-H transfer, in which CuBr and amino group are in the *trans* manner, is endergonic by 7.4 kcal mol^{-1} relative to **Int4**.

With the envelope conformation of pyrrolidine moiety in **Int5**, there are two axial hydrogen atoms (H^1 and H^2 in Figure 1) in the perfect orientation for transfer to the C^2 atom. The six- membered cyclic transition-state for the 1,5-H transfer has been located as **TS2_1** and **TS2_2** for the transfer of H^1 and H^2 atom, respectively. The free energy barrier for 1,5-H^1 transfer is calculated to be 27.7 kcal mol^{-1} (**TS2_1** relative to **Int4**), which is lower by 1.8 kcal mol^{-1} than that for the 1,5-H^2 transfer (**TS2_2**), due to the fact that H^1 is closer than H^2 to C^2

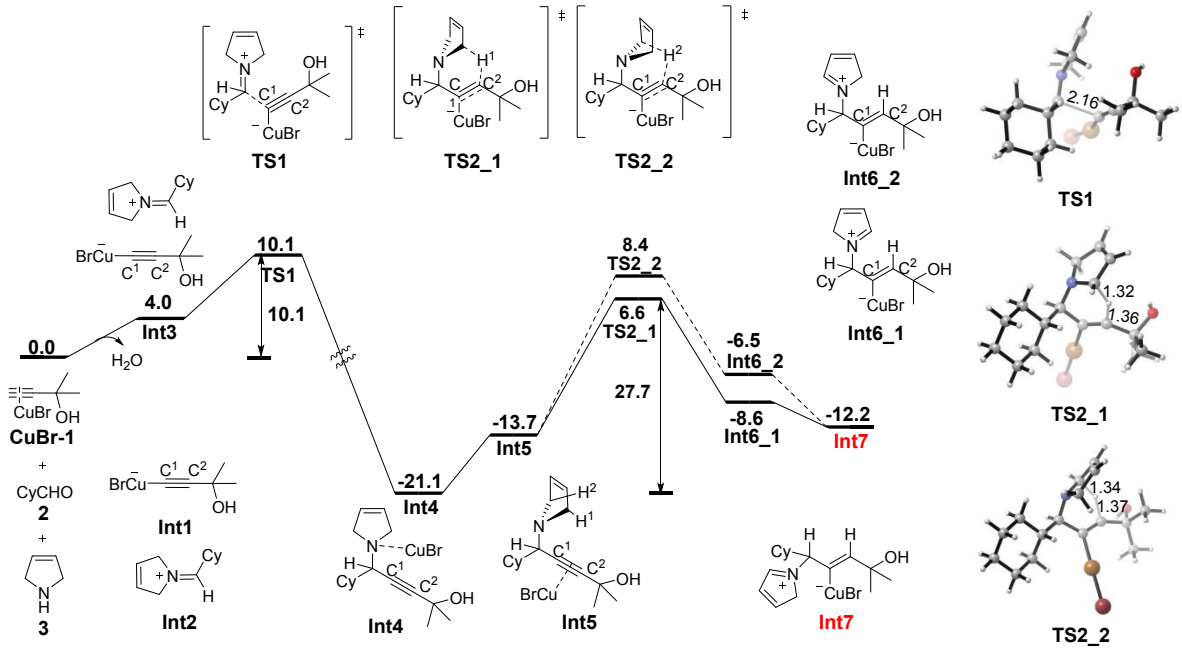


Figure 1. Free energy profile (ΔG in kcal mol^{-1}) for the formation of propargylic amine and the subsequent 1,5-H transfer. Bond lengths are given in angstroms

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atom (2.80 Å vs. 3.03 Å). Hence, **TS2_1** is more favorable, leading to the formation of *N*-allylic iminium-CuBr intermediate **Int6_1**, during which a σ bond between Cu(I) and C1 is formed. Subsequent isomerization of **Int6_1** by single-bond rotation produces the more stable species **Int7**, which is exergonic by 3.6 kcal mol-1 relative to **Int6_1**. The 1,5-H transfer step features the highest free energy barrier of the whole profile (27.7 kcal mol^{-1}, **TS2_1**), and may, thus, be rate-determining.

From iminium intermediate **Int7**, there are two alternative pathways: one is β-N elimination to produce allene product (*path a* in Figure 2), the other is protodemetalation process to afford the allylic amine (*path b* in Figure 2). The optimized transition structure for *path a* has been located as **TS3** and **TS4**, in which the C-N and C-Cu bonds are breaking simultaneously. The *cis* elimination from **Int7** proceeds through **TS3**, which requires an energy barrier of 24.9 kcal mol^{-1} (**TS3** relative to **Int7**). While the *trans* elimination through **TS4** is much easier with a barrier of 14.3 kcal mol^{-1}. The precursor for **TS4** is **Int8**, which is provided through the single-bond rotation of **Int7**. Thus, we consider that the β-N elimination should be in *trans* manner though **TS4**, which needs to overcome a free energy barrier of 14.3 kcal mol^{-1} to afford **Int9**. Finally, allene **5** is released from **Int9**, with an overall exergonicity of 19.2 kcal mol^{-1} (relative to **Int7**).

An alternative pathway from allyl iminium-CuBr intermediate **Int7** is the protodemetalation process to afford the allylic amine (*path b* in Figure 2), which is the only product obtained experimentally. The direct intramolecular protodemetalation of **Int7** proceeds through a concerted five-membered cyclic transition structure (**TS5**), in which the deprotonation of the alkyl hydrogen (H^3) adjacent to the N atom and protodemetalation of CuBr take place simultaneously, leading to the CuBr-coordinated allylic amine **Int11**. This process is computed to be exergonic (ΔG = 40.8 kcal mol^{-1}) and requires a 14.2 kcal mol^{-1} activation barrier (**TS5** relative to **Int7**). While **Int7** can isomerize to a less stable species **Int12** (1.0 kcal mol^{-1} higher in energy than **Int7**) through single-bond rotation with the hydroxyl group coordinating to Cu atom (d (Cu···O) = 2.41 Å). **Int12** may serve as the precursor for protodemetalation process as well, which also proceeds through a concerted five-membered cyclic transition structure (**TS6**). **TS6** needs to overcome a free energy barrier of 13.5 kcal mol^{-1}, which is lower in energy by 0.7 kcal mol^{-1} than **TS5** (14.2 kcal mol^{-1}), leading to **Int11** as well. Thus, we consider that **TS6** is preferable to **TS5** for the direct intramolecular protodemetalation of **Int7**. The final step of this process involves the release of the product allylic amine **4** through CuBr coordinated to another 3-pyrroline molecule, which is calculated to be exergonic by 10.4 kcal mol^{-1} (relative to **Int11**) due to the strong coordination ability of nitrogen with CuBr.

The above mentioned computational results show that the direct protodemetalation is preferred over β-N elimination both kinetically and thermodynamically, indicating that allyic amine should be provided as the major product. However, in the experiment, allylic amines are obtained as the only products, and allenes are not observed at all. We don't think that the free energy difference of 0.8 kcal mol^{-1} could account for the exclusive formation of allylic amine. In addition, deuterium-labeling experiments confirm that β-position proton of allylic amines may come from not only the azacycle but also the moisture in ambient environment (Scheme 2).[4] Hence, the stepwise protodemetalation assisted by other molecules is taken into account. When one molecule of water is taken into consideration,[14] the single bond C^2-C^3 in **Int7** rotates to facilitate the hydrogen bond forming between the hydroxyl group and the H^4 in water, leading to the formation of the complex **Int14**, in which three hydrogen bonds are taking shape with water [d (O^2···H^3) = 2.23 Å, d (O^1···H^4) = 1.96 Å, d (Br···H^5) = 2.73 Å] (Figure 3). The deprotonation process of **Int14** involves the abstraction

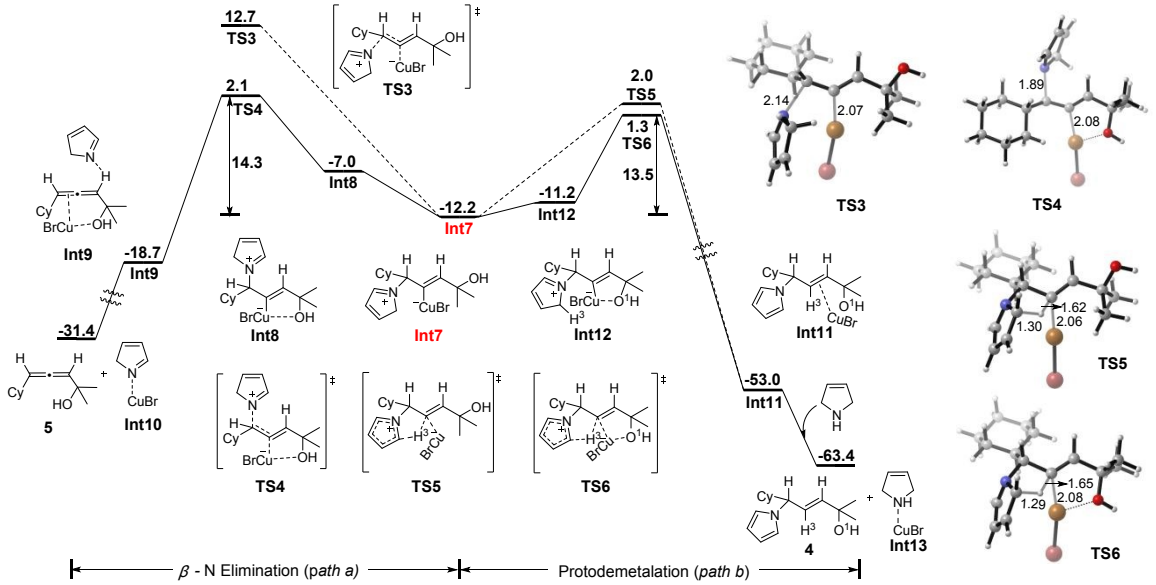


Figure 2. Free energy profile (ΔG in kcal mol^{-1}) for either intramolecular protodemetalation or β-N elimination from **Int7**. Bond lengths are given in angstroms.

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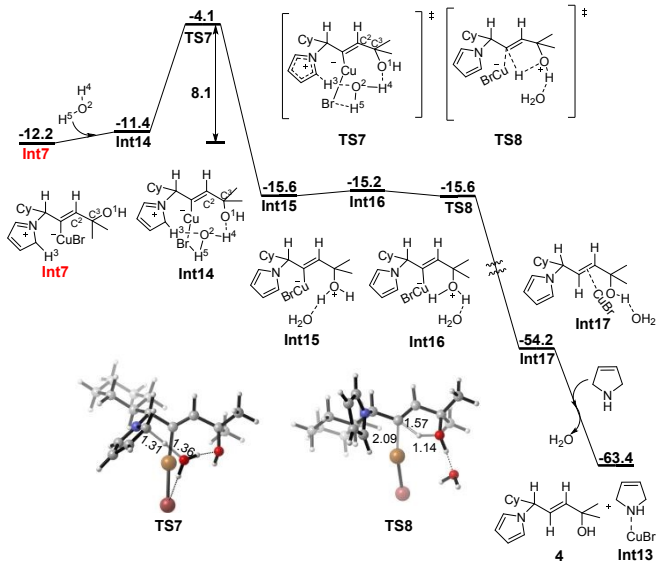


Figure 3. Free energy profile (Δ G in kcal mol^{-1}) for H_2O assisted stepwise protodemetalation of **Int7**. Bond lengths are given in angstroms

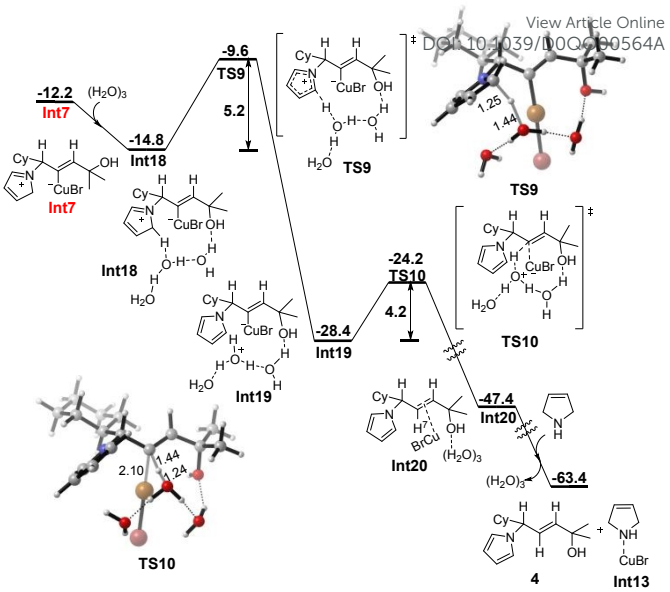


Figure 4. Free energy profile (Δ G in kcal mol^{-1}) for water cluster assisted stepwise protodemetalation of **Int7**. Bond lengths are given in angstroms

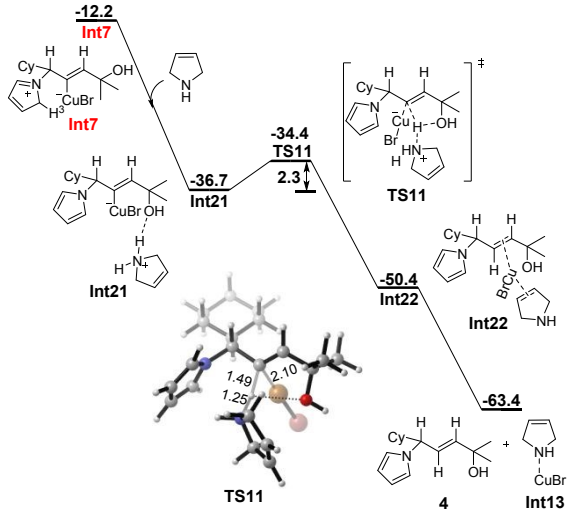


Figure 5. Free energy profile (Δ G in kcal mol^{-1}) for 3-pyrroline assisted stepwise protodemetalation of **Int7**. Bond lengths are given in angstroms.

of the proton H^3 from the azacycle by water and the subsequent protonation of the hydroxyl group, affording the intermediate **Int15**, in which the protonated hydroxyl group is stabilized by the hydrogen bond formed with water. This deprotonation process, accompanied by the aromatization of the azacycle of **Int14**, requires a free energy barrier of 8.1 kcal mol^{-1} (**TS7**, relative to **Int7**). Isomerization of **Int15** to **Int16** facilitates the subsequent protodemetalation process, which takes place intramolecularly through a concerted five-membered cyclic transition state (**TS8**). **TS8** is calculated to be higher in electronic energy by only 1.8 kcal mol^{-1} than **Int16**. However, when the thermal correction and the solvent effect are taken into account, **TS8** is slightly lower in free energy than intermediate **Int16**, indicating that the water-assisted protodemetalation could proceed with almost no barrier to afford the CuBr-coordinated allylic amine **Int17**. At last, the product allylic amine **4** can be obtained by the coordination of CuBr to another 3-pyrroline molecule.

When the water cluster, modeled by three molecules of water, is taking into account, a similar energy profile is obtained, which is showed in Figure 4. The water cluster assisted deprotonation requires to overcome a lower energy barrier of 5.2 kcal mol^{-1}. Different from one molecule water assisting pathway, the subsequent protodemetalation takes place in an intermolecular way with an energy barrier of 4.2 kcal mol^{-1}.

Calculations show that the presence of water (or water cluster) turns the direct concerted protodemetalation into a reversible stepwise deprotonation-protodemetalation process, which requires a much lower free energy barrier in total. Obviously, water (or the water cluster) may act as a proton shuttle in this process, making the protodemetalation process more kinetically favorable.

Interestingly, when an additional 3-pyrroline is taken into account as the proton shuttle, another kinetically favorable pathway of stepwise deprotonation-protodemetalation process has been discovered (Figure 5). All efforts to locate the hydrogen-bonding complex of 3-pyrroline with **Int7** are unsuccessful. When the N atom of 3-pyrroline is approaching the proton H^3 of azacycle of **Int7**, the abstraction of H^3 takes place immediately, indicating that the deprotonation process of **Int7** requires no barrier in the presence of 3-pyrroline. The abstraction of proton H^3 from **Int7** produces **Int21**, which is exergonic by 24.5 kcal mol^{-1}, accompanied by the aromatization of the azacycle. The subsequent protodemetalation takes place intermolecularly by the protonated 3-pyrroline through **TS11** with an energy barrier of 2.3 kcal mol^{-1}, affording the more stable CuBr-coordinated allyl amine **Int22**. Finally, the coordination of CuBr with 3-pyrroline releases allylic amine **4** to complete the whole process. Thus, an extra 3-pyrroline molecule could also act as the proton shuttle to change the concerted protodemetalation process to a stepwise deprotonation-protodemetalation process. But different from the water (or water cluster) involving pathway, the rate-limiting step of the 3-pyrroline assisting process is the protodemetalation step with

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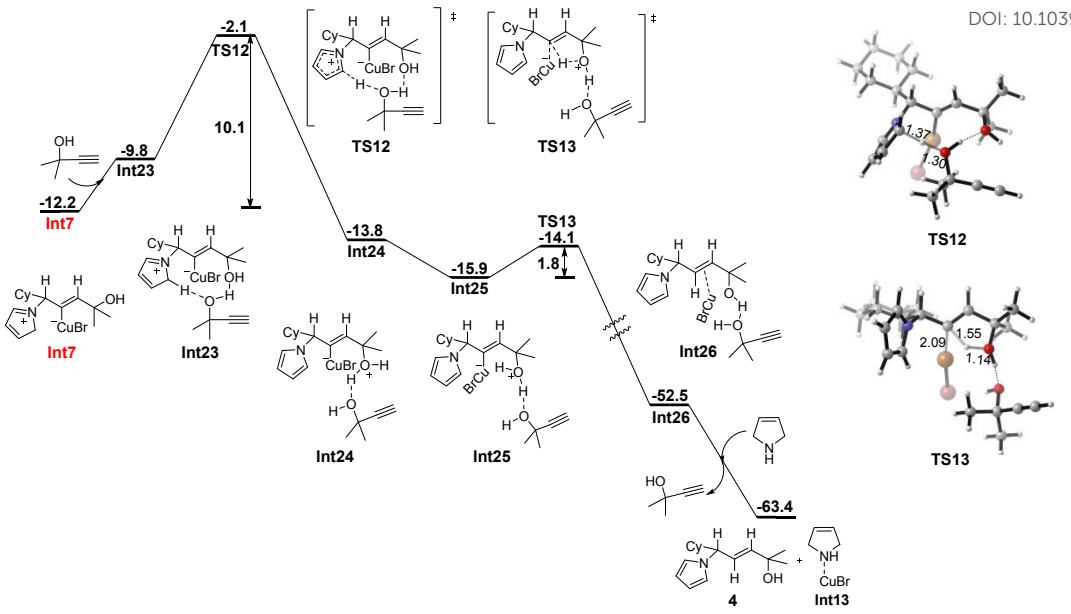


Figure 6. Free energy profile (ΔG in kcal mol^{-1}) for propargylic alcohol assisted stepwise protodemetalation of **Int7**. Bond lengths are given in angstroms.

the activation barrier of only 2.3 kcal mol^{-1}.

The propargylic alcohol **1**, as one of the reactants, may also act as a proton shuttle to assist the protodemetalation process (Figure 6). The computed potential energy surface is qualitatively similar to that of the one molecule water assisting pathway, but with more energy demanding in the deprotonation step, which needs to overcome an activation barrier of 10.1 kcal mol^{-1} (**TS12**). The following protodemetalation proceeds intramolecularly with an energy barrier of 1.8 kcal mol^{-1}.

The normal terminal alkynes with no hydroxy group present relatively lower reactivities in the experimental studies.[4] For example, when using 1-octyne under the standard reaction conditions, the *E*-allylic amine was formed as well with a lower yield (20% NMR yield), accompanied with 70% NMR yield of propargylic amine.[4] The effect of the hydroxyl group in 2-methybut-3-yn-2-ol (**1**) was further rationalized by DFT calculations on the comparison with the 1-octyne participated system.[15] The computed potential energy surfaces are qualitatively similar to that of 2-methybut-3-yn-2-ol (**1**) involved one, but more free energy demanding in the rate-limiting 1,5-H transfer step [28.8 kcal mol^{-1} (**TS15_1**) *vs*. 27.7 (**TS2_1**)], which is in accordance with the experimental observations. As shown in Figure 7, the hydrogen bond, formed between the hydroxyl group and the migrating hydrogen [*d* (O···H) = 2.51 Å] in **TS2_1**, facilitates the 1,5-H transfer, resulting in the enhancive stability of **TS2_1**.

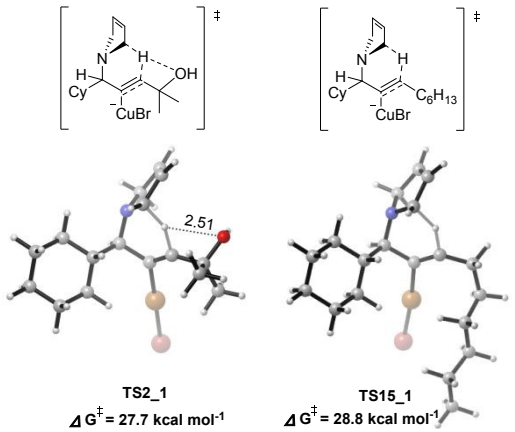


Figure 7. Optimized transition structures for 1,5-H transfer. Bond lengths are given in angstroms.

Conclusions

The activation energies for the stepwise deprotonation-protodemetalation process of **Int7** assisted by different species are listed separately in Table 1, together with the energy barrier of the β-N elimination process. It is obvious that the direct intramolecular pathway without the aid of other species is unfavorable, owing to the high energy demanding of 13.5 kcal mol^{-1}. The presence of different assisting species makes the concerted intramolecular process a irreversible stepwise deprotonation-protodemetalation process, reducing the activating barrier significantly. Among the four assisting species, 3-pyrroline shows the highest activity in deprotonation step owing to the relatively strong basicity of N atom. Once the rate-limiting 1,5-H transfer completes, the deprotonation-protodemetalation process will take place immediately with the aid of 3-pyrroline to afford the final allylic amine product. It is noteworthy that without 3-pyrroline (or isoindoline) in the reaction system, other species, such as water cluster, could act as the proton shuttle to assist the protodemetalation as well, which is in agreement with the observation of deuterium-labeling experiments (Scheme 2). Significantly, the aromatization of the azacycle is the essential driving force for the protodemetalation process.

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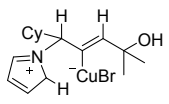
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Table 1. The summary of activation energies (ΔG in kcal mol^{-1}) for the processes from **Int7**.

Protodemetalation						β-N elimination
Assisting species	none[a]	H_2O	$(H_2O)_3$			
Deprotonation	13.5	8.1	5.2	No barrier	10.1	14.3
Protodemetalation		No barrier	4.2	2.3	1.8	

a: the protodemetalation proceeds in the concerted manner.



Int7

The pathway selectivity is determined by the activation energy difference between β-N elimination and the protodemetalation from the intermediate **Int7** formed *via* the 1,5-H transfer. Consistent with the experimental observations, with 3-pyrroline, the protodemetalation is favored over β-N elimination by 12.0 kcal mol^{-1} (14.3 kcal mol^{-1} *vs*. 2.3 kcal mol^{-1}), leading to the overwhelming formation of the allylic amine product. However, all the other amines listed in Scheme 1, other than 3-pyrroline and isoindoline, completely reverse the pathway selectivity, providing the allenes as the final products. Due to the lack of the driving force arising from the aromatization of azacycle, the protodemetalation of the intermediate **D** (Scheme 1) is rather difficult. Therefore, the β-N elimination becomes the favorable pathway for the classical ATA reactions using the amines except 3-pyrroline or isoindoline.

In summary, the mechanism of this Cu (I)-catalyzed reductive A^3-coupling reaction affording *E*-allylic amines was investigated by DFT calculations in detail. The calculations disclose that 3-pyrroline (or isoindoline) serves not only as a reactant, but also as a shuttle in a proton relay to assist the protodemetalation step, resulting in the production of allylic amine other than allene. Importantly, the calculations reveal the factors governing the selectivity: the aromatization of the azacycle (formed from 3-pyrroline or isoindoline) is essential driving force for the preference of the allylic amine products. Furthermore, we also discover that forming hydrogen bond with the migrating hydrogen to facilitate the rate-determining 1,5-H transfer step is the key role of the hydroxyl group in terminal alkyne substrate on the reactivity.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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15 All results for 1-octyne participated system, see Electronic Supplementary Information.

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